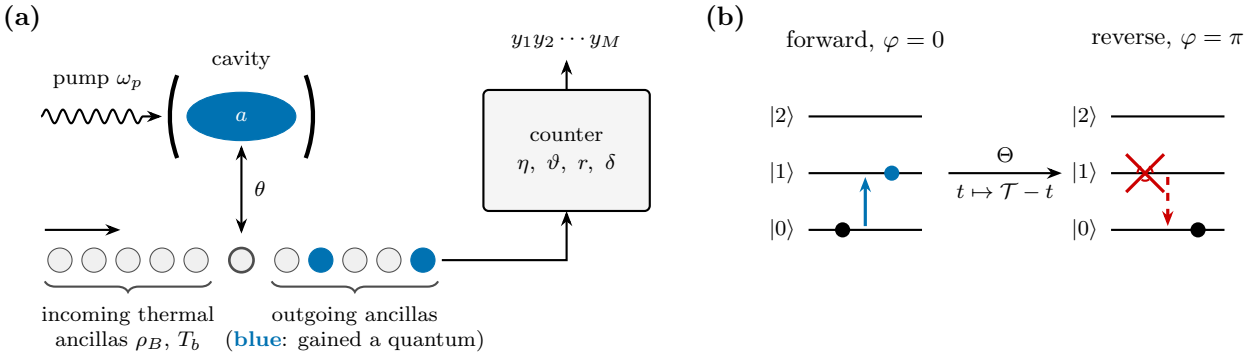

(a)
pump ω_p
cavity
a
θ
$y_1 y_2 \cdots y_M$
counter
$\eta,\ \vartheta,\ r,\ \delta$
incoming thermal
ancillas ρ_B, T_b
outgoing ancillas
(blue: gained a quantum)
(b)
forward, $\varphi = 0$
reverse, $\varphi = \pi$
$|2\rangle$
$|1\rangle$
$|0\rangle$
Θ
$t \mapsto \mathcal{T} - t$
$|2\rangle$
$|1\rangle$
$|0\rangle$